Lecture 11

Determinants

We shall given a recursive definition of determinants, that is, we shall define determinants of $n \times n$ matrices in terms of determinants of $(n-1) \times (n-1)$ matrices for $n \geq 2$ and define determinant of a 1×1 matrix (a_{11}) by $det(a_{11}) = a_{11}$.

Definition 25. *Let $n \geq 2$ and A be an $n \times n$ matrix.*

- *If we delete the i-th row and the j-th column of A, the determinant of the resulting $(n-1)\times(n-1)$ matrix is called the (i,j)-th minor of A and is denoted by M_{ij}.*
- *The number $(-1)^{i+j}M_{ij}$ is called the (i,j)-th cofactor of A and is denoted by C_{ij}.*
- *The determinant of A is defined by*

$$\begin{aligned} det(A) &= a_{11}C_{11} + a_{12}C_{12} + \ldots + a_{1n}C_{1n} \\ &= \sum_{j=1}^{n} a_{1j}C_{1j} \end{aligned}$$

We sometimes denote $det(A)$ by $|A|$ (which has nothing to do with absolute values).

Examples Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we have that:

$$M_{11} = d,\ M_{12} = c \text{ and } C_{11} = (-1)^2 d = d,\ C_{12} = (-1)^3 c = -c$$

and so $det(A) = aC_{11} + bC_{12} = ad - bc$ (which you have seen before).

Let $B = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 4 & 9 \\ 0 & 1 & 2 \end{pmatrix}$

$$\begin{aligned} \text{then } det(B) &= b_{11}C_{11} + b_{12}C_{12} + b_{13}C_{13} \\ &= 1\begin{vmatrix} 4 & 9 \\ 1 & 2 \end{vmatrix} - 2\begin{vmatrix} 0 & 9 \\ 0 & 2 \end{vmatrix} + 0\begin{vmatrix} 0 & 4 \\ 0 & 1 \end{vmatrix} \\ &= 1(4 \cdot 2 - 1 \cdot 9) - 2(0-0) + 0(0-0) \\ &= -1 \end{aligned}$$

Proposition 12. *Let $n \geq 2$ and A be an $n \times n$ matrix, then*

$$det(A) = a_{11}C_{11} + a_{21}C_{21} + \ldots + a_{n1}C_{n1} = \sum_{i=1}^{n} a_{i1}C_{i1}$$

The above says we can expand along the first column instead of the first row.

Example Given matrix B as in the above example, we could also expand along the first column and get:

$$\begin{aligned} det(B) &= b_{11}C_{11} + b_{21}C_{21} + b_{31}C_{31} \\ &= 1\begin{vmatrix} 4 & 9 \\ 1 & 2 \end{vmatrix} - 0\begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} + 0\begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} \\ &= 1(8-9) - 0(2-0) + 0(4-0) \\ &= -1 \end{aligned}$$

Corollary 1. *Let A be an $n \times n$ matrix, then $det(A^T) = det(A)$.*

Proposition 13. *Let A be an $n \times n$ and B the matrix obtained by interchanging any two distinct rows of A, then $det(B) = -det(A)$*

Corollary 2. *Let A be an $n \times n$ and B the matrix obtained by interchanging any two distinct columns of A, then $det(B) = -det(A)$*

Example $det \begin{pmatrix} 1 & 0 & 2 \\ 0 & 9 & 4 \\ 0 & 2 & 1 \end{pmatrix} = -(-1) = 1$ since this is determinant of our matrix B in first examples with column two and three interchanged.

Theorem 2. *Let $n \geq 2$ and A be an $n \times n$ matrix, then*

(a) $det(A) = \sum_{j=1}^{n} a_{ij}C_{ij}$ *for any $1 \leq i \leq n$, (we refer to this as expanding $det(A)$ along the i-th row).*

(b) $det(A) = \sum_{i=1}^{n} a_{ij}C_{ij}$ *for any $1 \leq j \leq n$, (we refer to this as expanding $det(A)$ along the j-th column).*

Proof. (a) Let $1 < i \leq n$. We can make row i the first row by interchanging adjacent rows $i - 1$ times. Let B be the matrix obtained this way,then we have that $det(B) = (-1)^{i-1}det(A)$, let A_{ij} denotes the $(n-1) \times (n-1)$ matrix obtained by deleting the i-row of A and j-th column of A and similarly for B, then it follows that $A_{ij} = B_{1j}$ and $a_{ij} = b_{1j}$ for $1 \leq j \leq n$.

$$\begin{aligned} det(A) = (-1)^{i-1}det(B) &= (-1)^{i-1} \sum_{j=1}^{n} (-1)^{1+j} b_{1j} det(B_{1j}) \\ &= (-1)^{i-1} \sum_{j=1}^{n} (-1)^{1+j} a_{ij} det(A_{ij}) \\ &= \sum_{j=1}^{n} (-1)^{i+j} a_{ij} det(A_{ij}) \\ &= \sum_{j=1}^{n} a_{ij} C_{ij} \end{aligned}$$

(b) Can be proved similarly (try it). □

Theorem 3. *Let A be an $n \times n$ matrix, then*

(a) If A is a triangular matrix (upper or lower), then $det(A) = a_{11}a_{22} \ldots a_{nn}$.
(b) If a row or column of A has only zeros, then $det(A) = 0$.
(c) If A has two identical rows (or columns), then $det(A) = 0$.

Proof. (a) We consider the sketch of the upper triangular case and leave the other as exercise. If A is upper triangular, the A has form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ 0 & 0 & \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

If we expand along the first column we get

$$det(A) = a_{11} \begin{vmatrix} a_{22} & a_{13} & \cdots & a_{1n} \\ 0 & a_{33} & \cdots & a_{2n} \\ 0 & 0 & \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix}$$

and we can continue expanding along the first column until we get to a_{nn}, thus getting our result.
(b) Use the above theorem expanding on the row or column with zeros.
(c) Interchanging the two identical rows (or columns) we get the same matrix A, but having interchanged the two rows we have $det(A) = -det(A)$ and so $det(A) = 0$. $\square$